

Early infarct FLAIR hyperintensity is associated with increased hemorrhagic transformation after thrombolysis

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Background and purpose: Absence of FLAIR hyperintensity within an acute infarct is associated with stroke onset <4.5 h. However, some patients rapidly develop FLAIR hyperintensity within this timeframe. We hypothesized that development of early infarct FLAIR hyperintensity would predict hemorrhagic transformation (HT) in patients treated with tissue plasminogen activator (tPA) < 4.5 h after onset.

Methods: Consecutive acute stroke patients treated with intravenous tPA <4.5 h after onset who had MRI before and 1 day after thrombolysis were included. Two raters (blind to HT) independently identified FLAIR hyperintensity with reference to the diffusion-weighted image (DWI) lesion. HT was assessed using T2* MRI at 24 h. Hemorrhagic infarction (HI) was defined as petechial HT without mass effect, and parenchymal hematoma (PH) as HT with mass effect. Multivariable logistic regression analysis for HT included FLAIR status, baseline National Institutes of Health Stroke Scale and DWI lesion volume, leukoaraiosis (Wahlund score), serum glucose and reperfusion.

Results: Of 109 patients, 33 (30%) had acute FLAIR hyperintensity. HT occurred in 17 patients (15.6%; 15 HI, 2 PH). HT was more common in FLAIR-positive patients than FLAIR-negative patients (33.3% vs. 9.2%, $P = 0.009$). Median time-to-scan and median time-to-thrombolysis did not differ significantly between patients with HT and without [97 IQR(68, 155) vs. 90 IQR(73, 119), $P = 0.5$; 120 IQR(99, 185) vs. 125 IQR(95, 150), $P = 0.6$, respectively]. In multivariable analysis, only FLAIR hyperintensity was independently associated with HT after thrombolysis (OR 18; 95% CI 2–175, $P = 0.013$).

Conclusions: Early development of FLAIR hyperintensity within the area of diffusion restriction is associated with increased risk of HT after thrombolysis in acute stroke patients.

Introduction

Fluid-attenuated inversion recovery (FLAIR) sequences are an integral part of stroke MRI and are useful to determine the time of symptom onset in acute ischemic stroke [1,2]. A mismatch between a positive diffusion-weighted image (DWI), in which the ischemic lesion is visible, and a negative FLAIR image, in which there are no parenchymal FLAIR hyperintensities, identifies patients highly likely to be within the therapeutic time

window [3]. However, it remains unclear whether hyperintensities on FLAIR sequences corresponding to areas of diffusion restriction on diffusion-weighted imaging (FLAIR positivity) can predict response to thrombolysis in acute ischemic stroke patients [4]. One study in 88 patients treated 0–6 h after stroke onset showed that FLAIR-positive patients had an increased risk of symptomatic intracerebral hemorrhage (sICH) following thrombolysis [5]. However, a more recent study in 49 patients showed that FLAIR positivity was not predictive of subsequent hemorrhage in patients treated 3–6 h after stroke onset [6]. The discrepancy of these results may be due to the different time windows in which patients were imaged.

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Hyperintensities on FLAIR may be the result of a net increase in water in pathological tissue following ischemia. A decline in cerebral blood flow leads to a failure of energy metabolism, resulting in cellular depolarization, an increase in tissue osmolality, and consequently a net increase of water. The extent and time course of this edema is dependent on the severity of ischemia [7]. In advanced stages of ischemia, disruption of the blood–brain barrier (BBB) leads to additional fluid influx into brain parenchyma. Hemorrhagic transformation (HT) is also believed to be the consequence of cytotoxic edema and breakdown of the BBB because of intense ischemia [8–10]. Therefore, early development of FLAIR hyperintensity within the standard therapeutic time window may indicate severe ischemia, cytotoxic edema, and increased BBB permeability with subsequent risk of HT.

In this study, we focused on patients imaged and treated with intravenous tissue plasminogen activator (tPA) within the standard 4.5 h treatment window. We sought to determine whether patients who rapidly develop FLAIR hyperintensity within this timeframe have a higher risk of HT after thrombolysis than FLAIR-negative patients.

Methods

Consecutive stroke patients recruited between May 2008 and May 2010 from two stroke centers (Berlin, local ethics committee EA4/026/08, and Hamburg, local ethics committee AZ M51/2000), with known time of onset, who had magnetic resonance imaging (MRI) before and 1 day after thrombolysis were included in this study after informed consent. In both centers, all patients were treated according to the current treatment guidelines; however, age limit was disregarded.

Minimum MRI protocol included the following sequences: diffusion-weighted imaging, perfusion imaging, T2*, and FLAIR; perfusion imaging and MR angiography were optional. Patients from Berlin were scanned on a 3T scanner (T2*: TE = 20 ms; TR = 620 ms; FOV = 220 mm; 5 mm slice thickness; 0.5 mm gap) and patients from Hamburg on a 1.5T scanner (T2*: TE = 23 ms; TR = 650 ms; FOV = 192 × 256 mm; 55 mm slice thickness; 0.5 mm gap). All patients received thrombolysis within 4.5 h. Follow-up MRI examinations were performed typically within 24 h after the pre-treatment scan [median time between scans in hh:mm was 23:34, interquartile range (IQR) 20:04–24:26].

Blinded to follow-up imaging, two raters (AK, MEBi) independently judged FLAIR positivity, defined as a hyperintensity on acute FLAIR images

corresponding (in shape and location) to the DWI lesion. Images were only rated FLAIR positive if both raters agreed on the presence of FLAIR hyperintensities corresponding to the area of diffusion restriction. Intensity and contrast windowing was limited to reflect standard image evaluation in the clinical setting. In the VINCI® MRI viewer (Max Planck Institute for Neurological Research, Cologne, Germany, Version 2.63) raters were only allowed to decrease %T[top] of DatMax to a minimum of 50% and increase %B[bottom] to a maximum of 10%. In the case of disagreement, the raters met for consensus. Perfusion deficit on MTT maps was defined as MTT > 6 s using Stroketool® (Digital Image Solutions, Frechen, Germany). Reperfusion was defined as a 75% reduction in perfusion deficit on follow-up compared with baseline [11]. Perfusion data were only available for Berlin patients (*n* = 90). HT was assessed on T2* sequences based on CT definitions previously published [12]: hemorrhagic infarction (HI) was defined as either petechial without mass effect (HI-1 for small petechiae, HI-2 for confluent petechiae), or parenchymal hematoma (PH) with mass effect occupying ≤30% of the infarcted area (PH-1) and occupying >30% (PH-2). sICH was defined as any bleed associated with clinical deterioration of ≥4 National Institutes of Health Stroke Scale (NIHSS) points [13]. Group comparison was made using Fisher's exact test and the Mann–Whitney *U*-test. Multivariable logistic regression analysis was performed to identify factors associated with HT. Along with FLAIR, we added known predictors of HT: NIHSS, DWI lesion volume, Wahlund score (a rating scale for age-related white matter changes) [14], serum glucose levels, and reperfusion [15].

Results

Of 109 patients (50.5% female; mean age, 71.3 years; standard deviation 12.5; median baseline DWI lesion volume 2.8 ml, IQR 0.23–9.3; 16 strokes in the posterior circulation), 30% were FLAIR positive (*n* = 33). Interrater agreement for FLAIR lesion hyperintensity was high (kappa = 0.78, *P* < 0.001). There were significantly more females in the FLAIR-negative group (58% vs. 33%, *P* = 0.02). FLAIR-positive patients had significantly higher median NIHSS scores on admission (10, IQR 6–7 vs. 6, IQR 4–11) and higher median baseline DWI lesion volumes (6.6 ml, IQR 2.4–20.3 vs. 1.1 ml, IQR 0.1–5.7, *P* < 0.01) than FLAIR-negative patients. For all baseline characteristics, see Table 1.

Seventeen patients (16%) had HT on the second scan (eight HI-1, seven HI-2, two PH-1, no PH-2:

Table 1 Baseline characteristics

	FLAIR positive (<i>n</i> = 33)	FLAIR negative (<i>n</i> = 76)	<i>P</i>
Age, mean (SD)	70 (12.3)	72 (12.6)	0.31
Gender, female, % (<i>n</i>)	33.3 (11)	57.9 (44)	0.02
Hypertension, % (<i>n</i>)	78.8 (26)	81.1 (60)	0.8
Glucose levels, median in mg/dl (IQR)	119 (112–185)	119 (107–142)	0.46
Diabetes mellitus, % (<i>n</i>)	30.3 (10)	20.3 (15)	0.32
Smoking, % (<i>n</i>)	30.3 (10)	18.9 (14)	0.22
Hyperlipidemia, % (<i>n</i>)	36.4 (12)	54.8 (40)	0.2
Atrial fibrillation, % (<i>n</i>)	33.3 (11)	29.7 (22)	0.82
Time-to-scan in minutes, median (IQR)	97 (75–145)	86.5 (71–119)	0.21
Time-to-thrombolysis in minutes, median (IQR)	120 (95–165)	124 (98–152)	0.86
Leukoaraiosis (Wahlund score), median (IQR)	5 (2–8)	5 (2–7)	0.9
Perfusion–diffusion mismatch, % (<i>n</i>)	52.2 (12)	62.3 (43)	0.46
Arterial occlusion, % (<i>n</i>)	54.5 (12)	49.3 (34)	0.81
Reperfusion, % (<i>n</i>)	37.5 (6)	36.7 (18)	1.0
Baseline NIHSS, median (IQR)	10 (6–17)	6 (4–11)	<0.1
Baseline DWI volume, median in ml (IQR)	6.6 (2.4–20.3)	1.1 (0.1–5.7)	<0.1
Baseline perfusion deficit, median in ml (IQR)	40.4 (13–123.5)	15.1 (2–60.3)	0.1

Fig. 1); one FLAIR-positive patient had an sICH (age 52, baseline DWI lesion volume 39 ml, initial blood pressure 115/60 mmHg). One FLAIR-negative patient had a microbleed unrelated to the DWI lesion on the first scan with no subsequent HT. FLAIR-positive patients had higher HT rates than FLAIR-negative patients [33.3% (six HI-1, 3 HI-2, and one PH-1) vs. 9.2% (two HI-1, four HI-2, and one PH-1), $P = 0.009$]. In a univariate analysis comparing patients with and without HT including all baseline parameters listed in Table 1, only FLAIR positivity was significantly different (56% vs. 25%, $P = 0.017$). Median time-to-scan and median time-to-thrombolysis did not differ significantly between patients with HT and without (97, IQR 68–155 vs. 90, IQR 73–119, $P = 0.5$;

Table 2 Multivariable regression analysis for HT

	Odds ratio	Confidence interval 95%	<i>P</i> -value
FLAIR	17.8	1.8–175.2	0.01
Baseline NIHSS	0.96	0.8–1.2	0.73
Baseline DWI lesion volume	1.1	0.9–1.2	0.29
Leukoaraiosis (Wahlund score)	1.1	0.8–1.6	0.62
Baseline serum glucose levels	1.0	0.9–1.0	0.66
Reperfusion	4.13	0.3–51.8	0.27

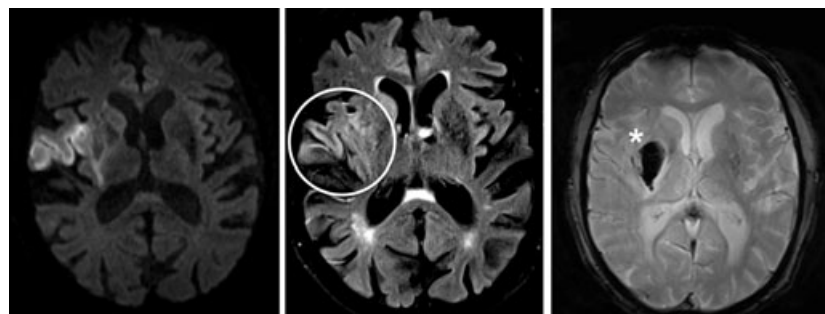
120, IQR 99–185 vs. 125, IQR 95–150, $P = 0.6$, respectively). Multivariable analysis identified FLAIR positivity as an independent predictor of HT (OR 18; 95% CI 2–175; $P = 0.01$; Table 2).

Discussion

This study suggests that early development of FLAIR hyperintensity is a risk factor for HT. Our results may help explain the apparent contradiction between previous studies on this subject. One study found an association between FLAIR positivity and risk of sICH in patients imaged up to 6 h after symptom onset [5]. The overall rate of FLAIR positivity was 30% – similar to our study – but the rate of FLAIR-positive patients in different sub-groups of onset time was not reported. Another recent study by Campbell and coworkers found no association between FLAIR positivity and HT [6]. In that study, patients were scanned 3 to 6 h post-stroke. This led to the almost universal presence of FLAIR positivity in their cohort making even the most elaborate analysis of association inconclusive.

Our study has demonstrated an association of FLAIR hyperintensity with HT in an earlier time window – in the timeframe for intravenous tPA. Patients who develop such early FLAIR hyperintensities may have more intense ischemia. In cases of good collateral flow, ischemia may be less severe, taking longer for these ischemic lesions to become FLAIR positive.

Figure 1 Example of a FLAIR-positive patient with subsequent hemorrhagic transformation (HT). Acute diffusion-weighted image is shown on the left, acute FLAIR image in the middle, T2* sequence on the right. FLAIR hyperintensity is encircled; HT is marked with an asterisk.



In line with this speculation, poor collateral flow and severity of ischemia have also been shown to increase risk of HT [8,9]. The proposed mechanism is cytotoxic edema, endothelial injury, and BBB permeability [10]. In this light, HI and PH can be seen as part of the same pathophysiological spectrum of ischemic injury, despite their very different clinical implications. While most authors have found that HI has no detrimental effect on outcome [16] and may even be a marker of early reperfusion and favorable outcome [17], PH is usually associated with poor clinical outcome, regardless of whether it meets the criteria for sICH [18].

FLAIR is known to be a good indicator of time of onset [1,19,20], and time-to-thrombolysis may play a role in increased risk of HT [18], although only a non-significant trend has been demonstrated in meta-analysis of thrombolysis trials [21]. However, we found no difference in time-to-scan or time-to-thrombolysis between patients with and without HT, or between FLAIR-positive and FLAIR-negative patients within this 0–4.5 h time window. Similar to previous studies, we found that FLAIR positivity is associated with larger DWI lesions [1,6,19]. However, in our analysis, FLAIR positivity was a significant independent predictor of HT. Baseline infarct volume is a known predictor of HT as well [22]; the small median baseline DWI lesion volume of this cohort most probably explains why this parameter did not turn out to be significant in the multivariable regression analysis (Table 2). We have no explanation for the gender differences between FLAIR-positive and FLAIR-negative patients; the addition of gender into a multivariable analysis for HT did not change our conclusions.

Whilst we have demonstrated a potential pathophysiological connection between FLAIR positivity and HT, the clinical implications of this study are less clear. Only one sICH occurred in these 109 patients, which is consistent with the rates in current trials and registries of thrombolysis within 4.5 h [23,24]. Given the evident safety of thrombolysis using current criteria, we would not advise clinicians to exclude patients with acute FLAIR hyperintensity from thrombolysis within the 4.5 h time window.

This is a retrospective, exploratory study with several limitations. We cannot rule out that some of our FLAIR-positive patients had recent silent ischemia (RSI) on acute FLAIR. But others have shown that RSI was not associated with increased risk of HT [25]. Additionally, we used an arbitrary definition of FLAIR positivity. There is an on-going debate on how to standardize the criteria for the evaluation of FLAIR [26]. Although quantitative FLAIR analysis (using automated signal-intensity thresholds) maximizes specificity [20] and may com-

pensate for the subjectivity of pure visual analysis, this method is time-consuming and may delay the decision to treat. Recent studies have confirmed the high specificity, sensitivity, and interrater reliability of qualitative FLAIR analysis [2,20]. Qualitative image assessment is therefore an accurate and accessible method and may be more appropriate in the clinical setting.

In conclusion, in addition to being a surrogate marker of lesion age, FLAIR sequences may be a useful tool to assess the likelihood of subsequent HT in patients treated with intravenous tPA within 4.5 h of symptom onset. Nonetheless, given the safety of thrombolysis within this time window, patients should not be excluded from thrombolytic treatment based on this phenomenon.

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